

IOWA FALLS, IA, USA

WMO: 722331

Lat: 42.467N

Lon: 93.267W

Elev: 1137

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 54941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -4.2	(d) -15.2	(e) 2.5	(f) -8.6	(g) -10.5	(h) 3.3	(i) -3.8	(j) 34.3	(k) 17.8	(l) 31.9	(m) 15.8	(n) 8.8	(o) 290	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 90.8	(d) 74.3	(e) 87.9	(f) 73.2	(g) 84.7	(h) 71.3	(i) 77.3	(j) 86.6	(k) 75.4	(l) 83.8	(m) 73.8	(n) 82.0	(o) 11.7	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	135.1	83.0	72.8	126.8	80.8	71.3	120.3	79.4	41.6	86.7	39.7	84.2	38.2	82.0	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	25.1	22.0	-19.0	95.2	4.9	3.7	-22.5	97.8	-25.3	100.0	-28.1	102.0	-31.6	104.7
			-19.3	80.3	4.6	1.8	-22.6	81.5	-25.3	82.5	-27.8	83.5	-31.1	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	18.0	21.6	35.1	48.0	60.2	70.5	72.3	69.6	64.1	50.4	36.5	24.1
	DBStd	21.53	12.94	12.40	12.40	9.49	9.09	6.23	5.96	4.95	7.97	9.10	10.55	11.41
	HDD50	3771	992	796	482	148	18	0	0	3	110	418	804	
	HDD65	7055	1457	1216	927	511	206	20	10	13	111	459	856	1269
	CDD50	2917	0	1	21	89	333	615	691	608	426	121	12	0
	CDD65	724	0	0	1	2	55	184	236	157	84	5	0	0
	CDH74	6577	0	0	13	62	553	1679	2169	1257	768	75	1	0
	CDH80	2081	0	0	0	7	172	593	744	322	232	11	0	0
	WSAvg	9.7	11.7	11.6	11.4	12.3	11.0	8.6	6.2	5.3	7.0	9.1	11.0	11.0
	PrecAvg	33.8	0.9	1.0	2.0	3.3	4.2	5.0	4.4	4.1	3.3	2.4	1.8	1.1
Precipitation	PrecMax	53.6	3.1	3.1	5.0	7.9	9.6	15.4	9.7	13.6	13.3	5.2	5.6	4.6
	PrecMin	22.4	0.0	0.0	0.1	0.7	0.9	1.7	0.7	0.4	0.7	0.2	0.0	0.2
	PrecStd	6.9	0.6	0.7	1.2	1.7	2.0	2.6	2.3	2.4	2.1	1.5	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	48.1	58.4	75.1	80.6	91.3	93.4	93.5	90.6	90.5	81.1	69.8	54.9	
	0.4% MCWB	42.5	51.6	60.1	61.6	69.7	74.0	77.0	74.5	73.5	62.5	56.8	46.9	
	2% DB	41.0	46.4	65.7	74.8	84.4	90.3	90.7	86.5	86.4	75.1	61.4	46.9	
	2% MCWB	37.4	42.3	55.4	58.7	66.2	73.0	76.3	73.9	71.3	60.6	53.0	43.3	
	5% DB	37.1	41.6	59.0	69.6	80.5	86.4	87.3	83.7	82.0	70.5	55.6	42.8	
	5% MCWB	35.1	38.1	51.6	55.0	64.0	70.8	75.5	72.7	68.3	59.0	48.4	39.4	
	10% DB	34.6	37.2	52.1	64.3	75.6	83.1	84.0	80.9	78.5	66.1	51.7	37.5	
	10% MCWB	33.2	35.0	46.5	53.1	62.8	68.7	72.6	69.9	66.2	57.1	45.7	35.2	
	0.4% WB	43.5	52.7	62.4	64.0	72.0	79.4	80.3	78.0	76.6	68.4	58.6	50.5	
	0.4% MCDB	48.7	59.9	72.6	75.7	84.6	89.7	89.4	87.4	87.0	73.8	65.1	53.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	37.7	43.7	57.3	60.7	69.4	75.9	78.0	75.4	73.6	64.7	54.0	44.0	
	2% MCDB	40.0	46.2	63.4	70.1	81.4	85.0	86.9	83.2	82.0	70.1	61.4	46.1	
	5% WB	35.3	38.7	51.8	58.2	67.3	73.7	76.1	73.8	71.3	61.6	49.6	39.0	
	5% MCDB	37.2	41.5	58.1	67.4	76.3	82.6	84.7	81.1	79.0	67.6	54.4	42.7	
	10% WB	33.3	35.0	46.5	54.8	65.2	71.6	74.3	72.1	69.1	58.0	45.8	35.5	
	10% MCDB	34.4	37.2	51.9	62.7	72.8	80.3	82.6	78.7	75.3	65.1	51.2	37.6	
Mean Daily Temperature Range	MDBR	15.3	16.3	16.5	20.5	20.5	19.5	19.8	20.1	22.5	21.1	18.2	14.5	
	5% DB MCDBR	16.1	18.8	27.2	29.8	25.8	23.4	21.8	21.8	25.0	27.4	25.8	18.8	
	5% DB MCWBR	13.9	15.0	17.4	16.1	11.3	10.3	10.8	11.1	11.8	14.8	16.9	14.5	
	5% WB MCDBR	15.9	18.4	25.6	25.0	21.1	20.7	20.2	19.2	20.8	21.5	24.5	18.0	
	5% WB MCWBR	14.0	15.0	16.9	15.6	11.0	11.3	10.9	10.8	11.1	14.3	17.4	14.4	
Clear-Sky Solar Irradiance	taub	0.278	0.295	0.323	0.371	0.400	0.418	0.431	0.412	0.383	0.338	0.299	0.276	
	taud	2.413	2.370	2.403	2.302	2.287	2.285	2.225	2.285	2.325	2.447	2.489	2.438	
	Ebn at Noon	277	289	293	284	278	272	266	268	268	269	267	267	
	Edh at Noon	24	30	33	39	41	42	44	40	36	28	23	22	
All-Sky Solar Radiation	RadAvg	590	914	1205	1495	1771	1964	2004	1724	1417	938	633	474	
	RadStd	63	72	121	133	129	165	117	115	99	120	50	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page